Bazaar Inventory Tracking System

Case Study Challenge | Engineering

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بازار

BAZAAR

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shaping futures

Overview

This system tracks product inventory and stock movements across multiple stores as given in the Case Study PDF, with support for real-time stock visibility, audit capabilities, and scalable architecture to handle thousands of stores.

Features

- **Track inventory:** Monitor stock levels across stores in real-time
- **Stock movements:** Record stock-in, sales, and manual removals
- **Multi-store support:** Central product catalog with store-specific inventory
- **Authentication & authorization:** Role-based access control with store-specific permissions
- **Reporting:** Low stock alerts, inventory valuation, and sales tracking
- **Audit:** Complete history of all inventory changes
- **Scalability:** Designed to handle thousands of stores and high transaction volumes

Bazaar Inventory Management System - v1

Stage 1: Single Store

SQLite Database: Local storage for a single store

- Embedded relational database stored on disk (inventory_stage1.db)
- Manages product information, inventory levels, and stock movement history
- Ideal for lightweight, single-user environments

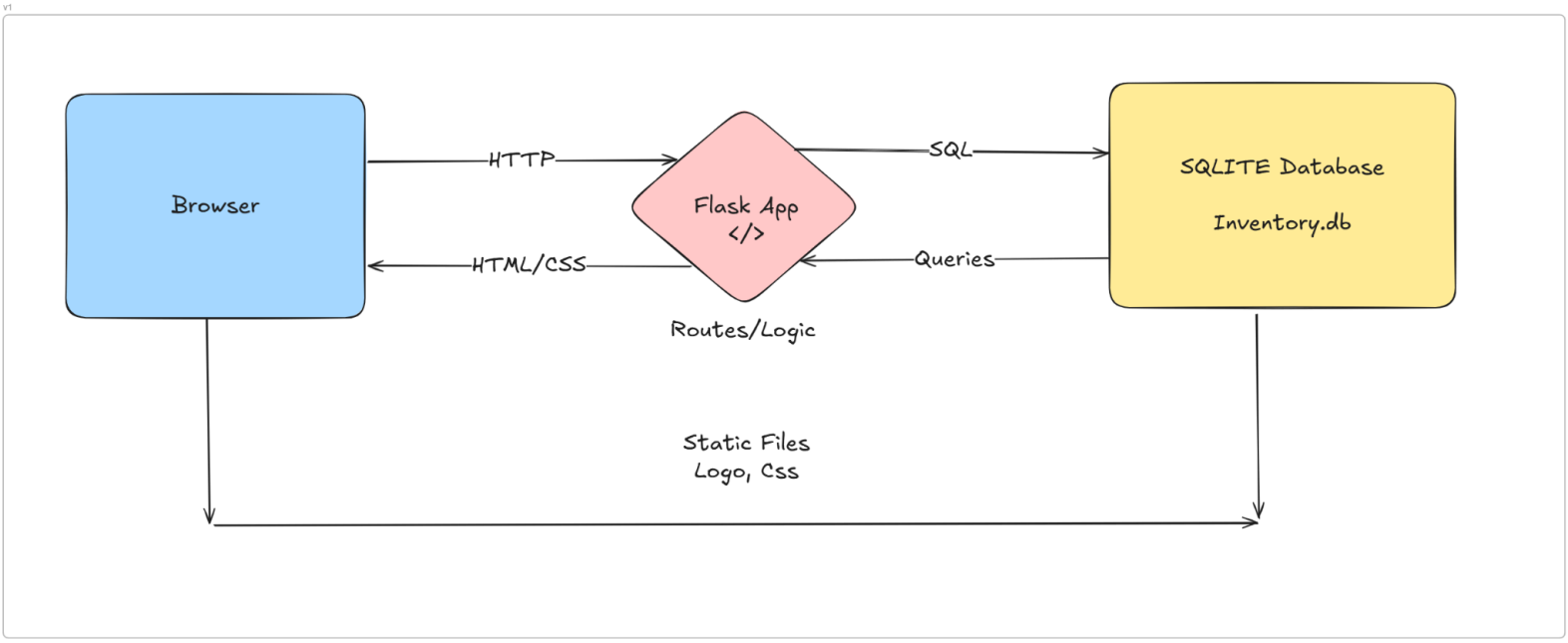
Backend On Flask

- Implemented using Python (Flask)
- Adding and retrieving products
- Recording inventory actions (stock-in, sale, removal)
- Responsible for database initialization, validation, and transaction integrity

User Interface

- Web based interface for easy-to-understand UI
- User-friendly forms for product management
- Visual dashboard for live inventory overview
- Bootstrap-enhanced for responsiveness

System Architecture Diagram for v1:



Bazaar Inventory Management System - v2

Design Decisions

Multi-Store Architecture

- Central Product Catalog: All stores share a common product database, enabling consistent product information across the organization
- Store-Specific Inventory: Each store maintains separate inventory records, allowing for location-specific stock management
- Event-Based Stock Movement: All inventory changes are tracked as events with timestamps, users, and references for complete audit trail

Database Design

- Relational Database: Chose PostgreSQL for ACID compliance, transaction support, and better concurrency handling
- Normalization: Database schema follows proper normalization principles to avoid data redundancy
- Foreign Key Constraints: Enforces data integrity across related tables
- Timestamp Tracking: All entities include created_at and updated_at fields for audit and tracking

Security Implementation

- JWT Authentication: Secure, stateless authentication using JSON Web Tokens
- Role-Based Access Control: Three primary roles - admin, store_manager, and store_user

- Store-Based Permissions: Users have access only to their assigned stores (except admins)
- Password Hashing: Passwords are securely hashed and never stored in plain text
- Rate Limiting: API endpoints are protected against brute force and DoS attacks

Containerization Strategy

- Docker Isolation: Each component runs in its own container for better isolation and scalability
- Simplified Deployment: Docker Compose orchestration simplifies setup across environments
- Environment Consistency: Ensures consistent behavior across development and production
- Volume Persistence: Database data persists across container restarts

Key Assumptions

- Internet Connectivity: Assumes reliable internet connectivity for API communications between stores
- Data Volume: System can handle moderate transaction volumes (up to thousands per day per store)
- Concurrency: Multiple users may access the system simultaneously from different stores
- Transaction Size: Most inventory operations involve reasonable quantities (not millions of items in a single transaction)
- Product Uniqueness: Products are unique across the entire organization with a single central catalog

- User Access Patterns: Store users primarily access their own store's data, while admins need cross-store visibility

API Design

Design Principles

- RESTful Architecture: Clear resource-oriented design with appropriate HTTP methods
- Versioning: All endpoints prefixed with /api/v2/ to allow future version changes
- Consistent Response Format: Standardized success and error response structures
- Pagination: All list endpoints support pagination to handle large data sets
- Filtering: Flexible query parameters for filtering data
- Authentication: JWT-based with refresh token capability
- Documentation: Comprehensive API documentation with examples

Endpoint Structure

- Authentication Endpoints: /api/v2/auth/* for login, registration, token refresh
- Product Endpoints: /api/v2/products/* for central catalog management
- Inventory Endpoints: /api/v2/inventory/* for stock management
- Store Endpoints: /api/v2/stores/* for store management
- Reporting Endpoints: /api/v2/reports/* for analytics and insights

Security Measures

- Input Validation: All request data is validated before processing
- Rate Limiting: Prevents API abuse with per-endpoint limits
- Authorization Middleware: Enforces access control at the route level
- Secure Headers: Properly configured security headers

Evolution Rationale (v1 → v2)

Architectural Evolution

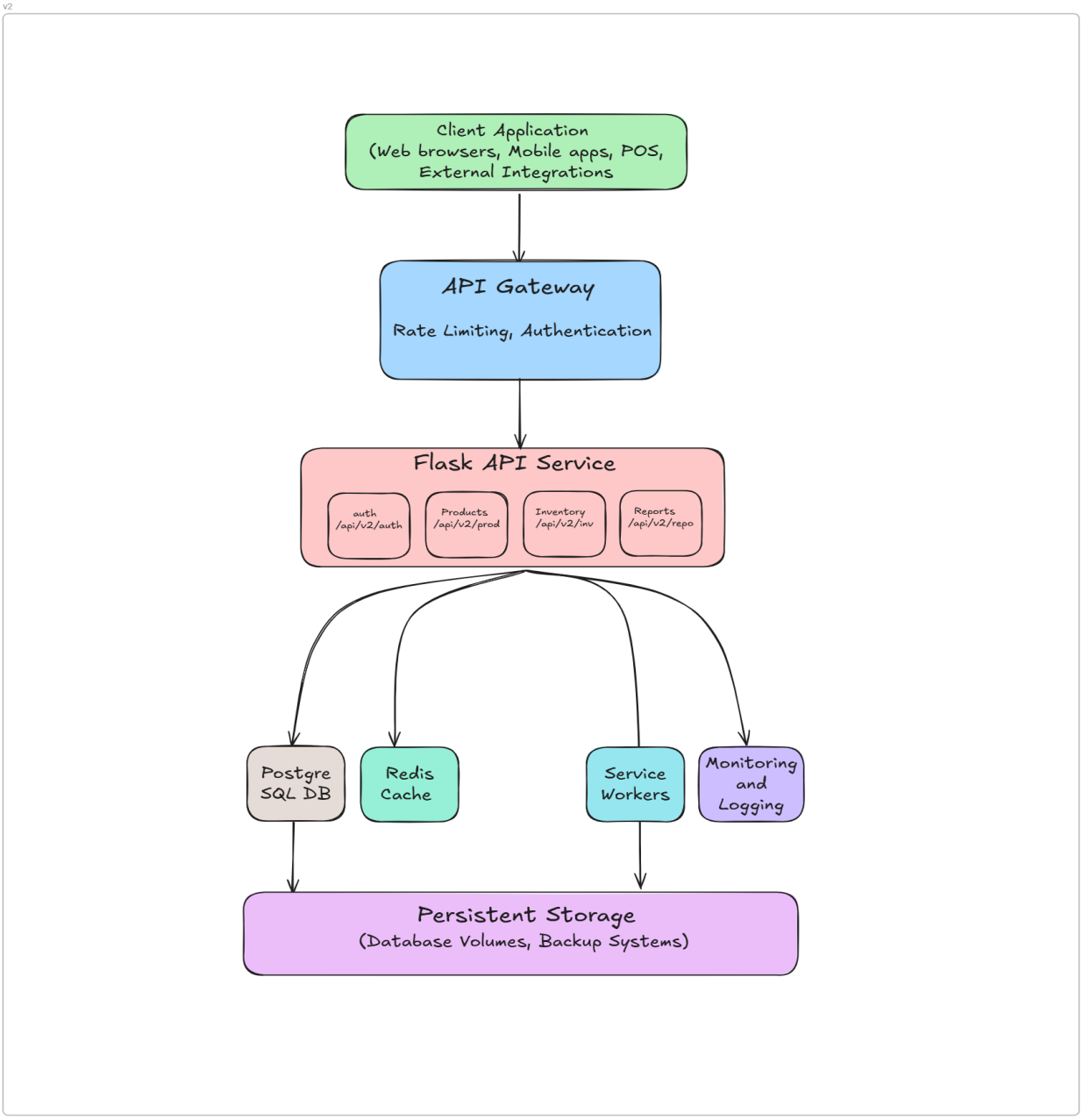
- From Monolith to API Service:
    - v1: Single-application monolith with direct HTML rendering
    - v2: API-first design with separation of concerns, enabling multiple front-end options
- From Single to Multi-Store:
    - v1: Limited to a single store's inventory
    - v2: Supports multiple stores sharing a central product catalog
- From Basic to Advanced Security:
    - v1: Simple session-based authentication
    - v2: JWT-based authentication with role-based access control
- From File-based to Relational Database:
    - v1: SQLite for simplicity
    - v2: PostgreSQL for scalability and concurrency

Technical Implementation Changes

- Database Migration:
    - Enhanced schema with store-related tables
    - Added foreign key relationships
    - Improved indexing strategy

- Authentication System:
    - Implemented JWT token-based authentication
    - Added token refresh mechanism
    - Created role-based authorization
- API Structure:
    - Organized endpoints into logical blueprints
    - Added versioning support
    - Implemented consistent response formatting
- Containerization:
    - Added Docker and Docker Compose
    - Created separate containers for API, database, and cache
    - Implemented volume persistence
- Performance Enhancements:
    - Added Redis for rate limiting and caching
    - Implemented database connection pooling
    - Optimized query patterns

System Architecture Diagram for v2:



Bazaar Inventory Management System v3

Design Decisions

Microservices Architecture

- Separated core functionality into independent components (API service, worker service)
- Used message queues for communication between services
- Implemented containerization for all components to enable independent scaling

Database Read/Write Separation

- Implemented primary/replica architecture to optimize performance
- Used a session manager to route read and write operations appropriately
- Added failover capability to maintain availability during replica downtime

Caching Strategy

- Implemented Redis-based caching for frequently accessed data
- Applied cache invalidation patterns to maintain data consistency
- Created cache management endpoints for administration

API Rate Limiting

- Implemented rate limiting based on client IP
- Used different rate limits for various endpoints based on impact
- Stored rate limit data in Redis for persistence across API instances

Load Balancing

- Incorporated Nginx as a reverse proxy and load balancer
- Implemented health checks to ensure traffic only goes to healthy instances
- Used least connections algorithm for optimal request distribution

Assumptions

Security Assumptions

- JWT tokens are sufficient for authentication and authorization
- Access control by store is required for multi-store environment
- Admin users need system-wide access to all stores

Data Assumptions

- Products have a central catalog shared across all stores
- Inventory is store-specific
- Stock movements must be trackable and auditable

Scaling Assumptions

- Read operations will significantly outnumber write operations
- Report generation may be resource-intensive and should be processed asynchronously
- The system must handle thousands of stores with minimal performance degradation

User Assumptions

- Multiple user roles are needed (admin, store manager, store user)
- Users will primarily access the API through a frontend application
- API will need to be consumed by both web and mobile clients

API Design

RESTful Principles

- Used consistent resource-based URL structure
- Implemented proper HTTP methods (GET, POST, PUT, DELETE)
- Applied consistent response formatting with appropriate status codes

Versioning Strategy

- Implemented URL-based versioning (v2 → v3)
- Maintained backward compatibility where possible
- Added new endpoints for enhanced functionality

Authentication and Authorization

- JWT-based authentication with separate login endpoint
- Role-based access control for different operations
- Store-specific access restrictions for non-admin users

Pagination and Filtering

- Implemented standard pagination for list endpoints
- Added filtering capabilities for complex data queries
- Included metadata in responses for client-side pagination handling

Evolution Rationale (v2 → v3)

Scalability Improvements

- Added horizontal scaling capabilities for all services

- Implemented database read/write separation for better performance
- Containerized all components with Docker Compose for easier deployment and scaling

Reliability Enhancements

- Added asynchronous processing via message queues
- Implemented background workers for resource-intensive operations
- Added health checks and failover mechanisms

Performance Optimizations

- Implemented Redis caching for frequently accessed data
- Added database connection pooling
- Optimized database queries with proper indexing

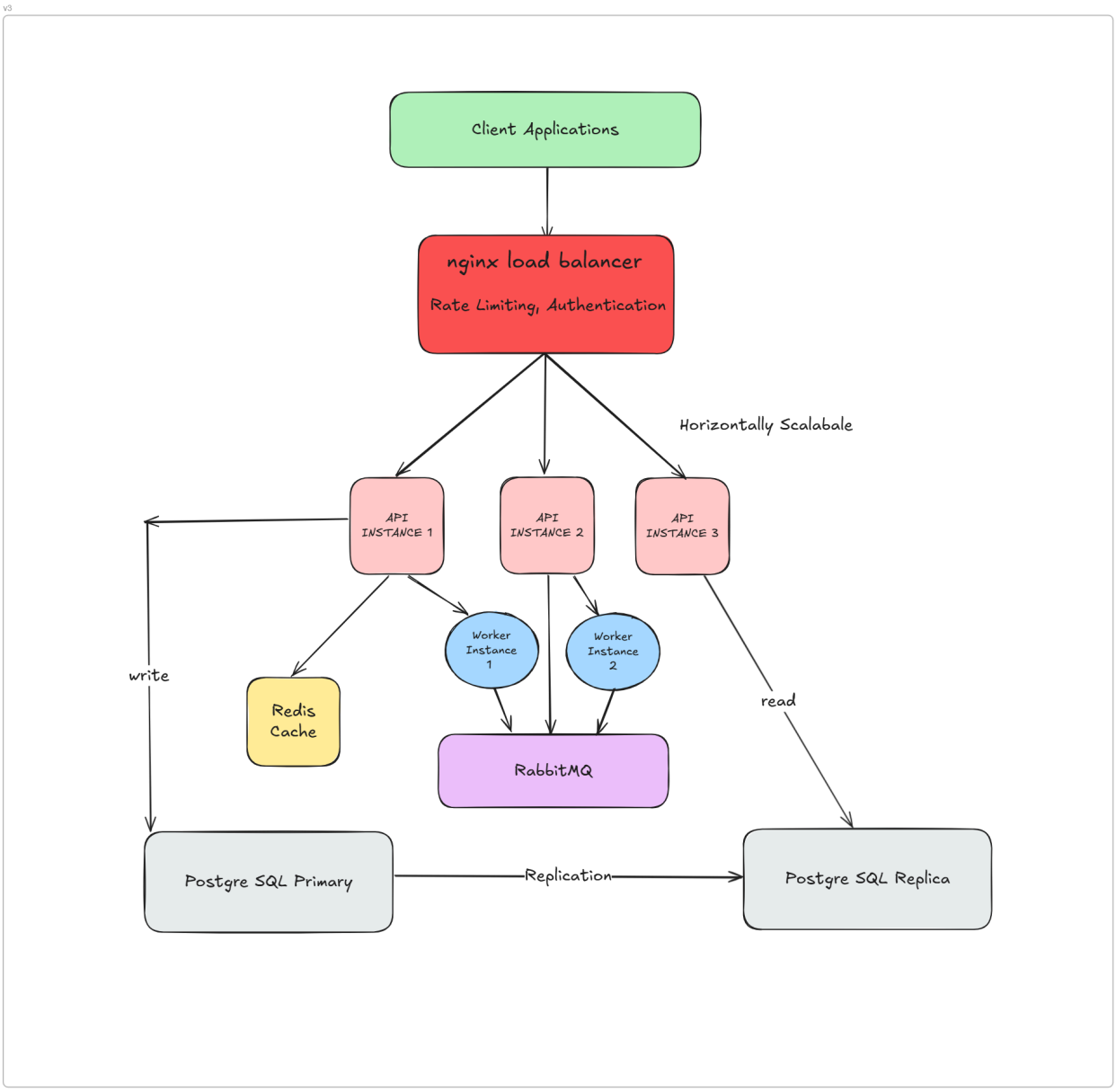
Architecture Modernization

- Moved from monolithic design to microservices
- Added event-driven communication between services
- Implemented infrastructure as code for consistent deployment

Advanced Features

- Added comprehensive reporting capabilities
- Implemented asynchronous report generation
- Enhanced audit logging for all inventory movements

System Architecture Diagram for v3:



Please find the [Github Repository] for complete code and further tests.